

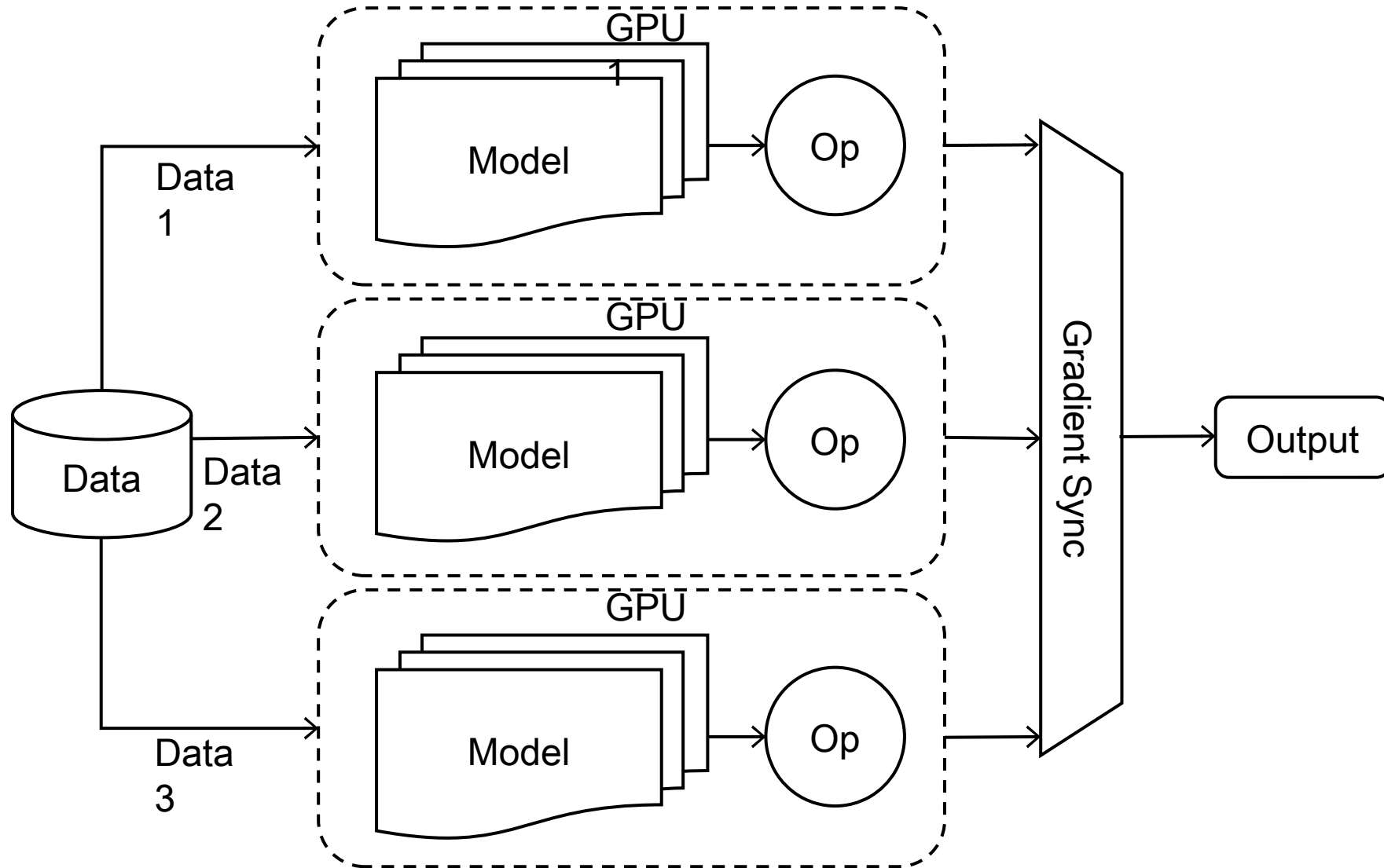
Update – Sep 23

JC Vaught

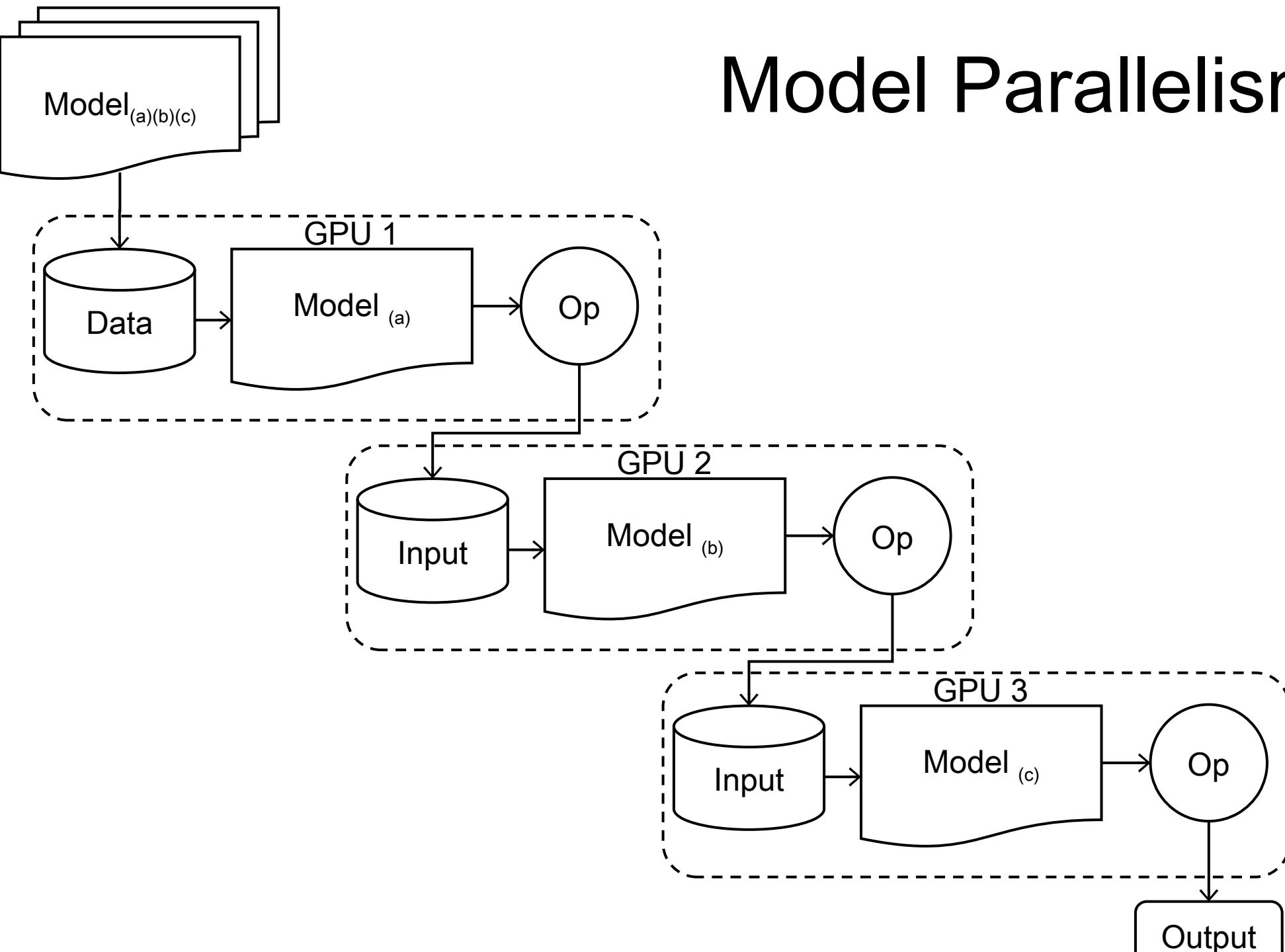
GPU Sharding

- Splitting workload onto multiple GPUs.
- 4 methods
 - Data Parallel - easy
 - Model Parallel - medium
 - Pipeline Parallel - harder
 - Tensor parallel – harder
 - Hybrid - Hardest
- Why use each method?
 - Data Parallel – speed up training - small
 - Model Parallel – Train Deep models
 - Pipeline Parallel – Deep Models, efficient
 - Tensor parallel – Wide Models
 - Hybrid – Can be tweaked as needed
 - May not result in much improvement results

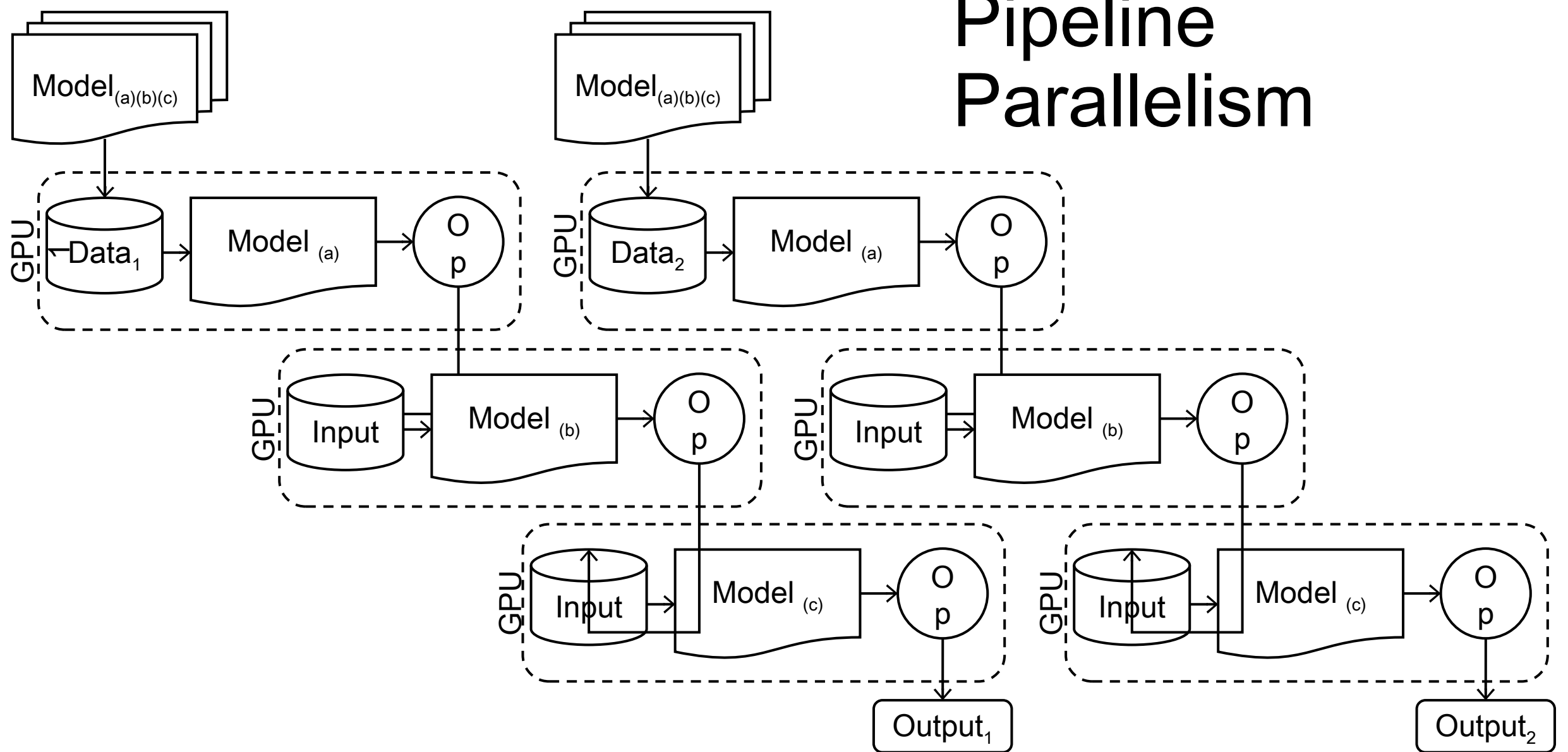
Data Parallelism



Model Parallelism



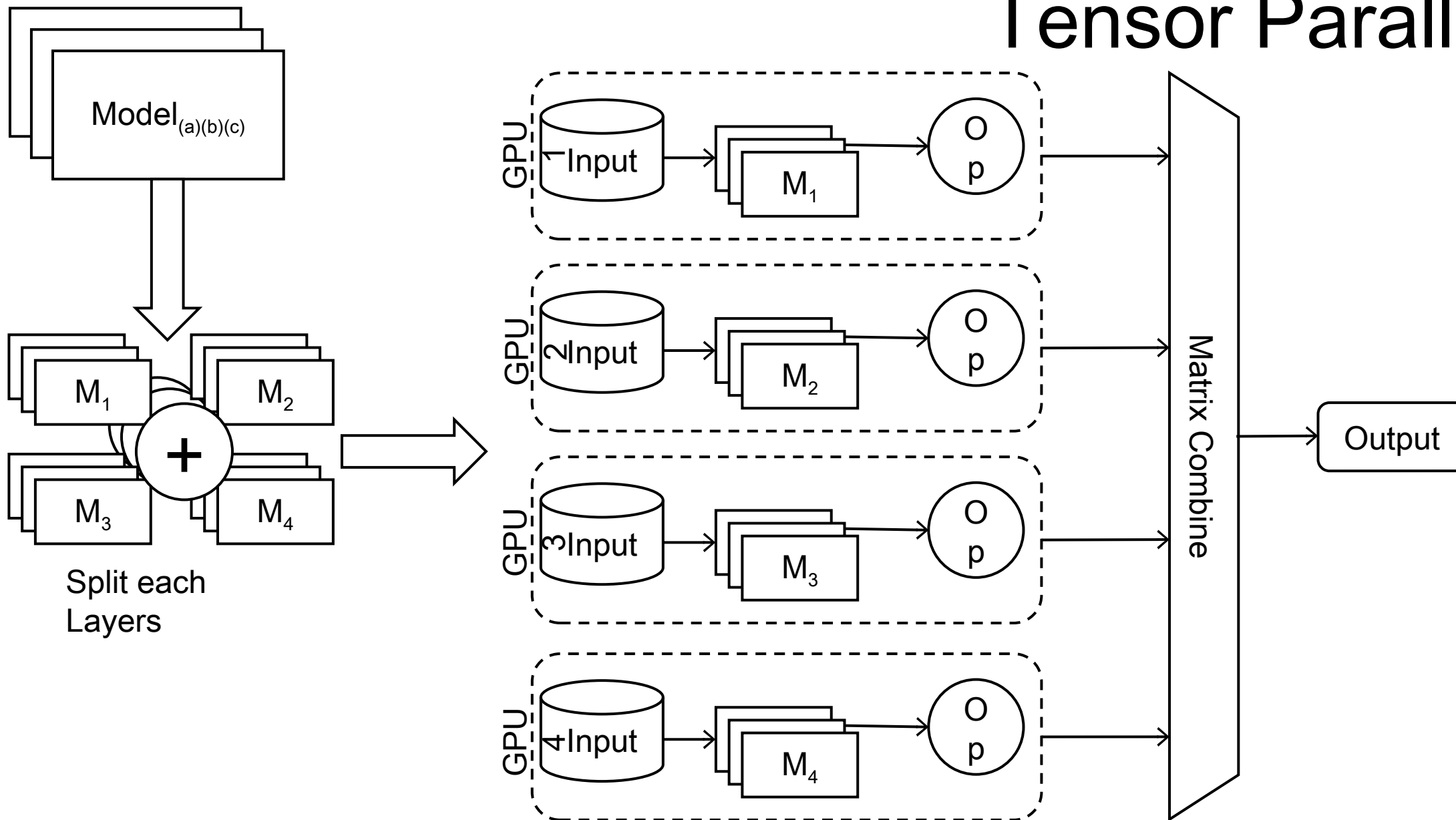
Pipeline Parallelism



Time
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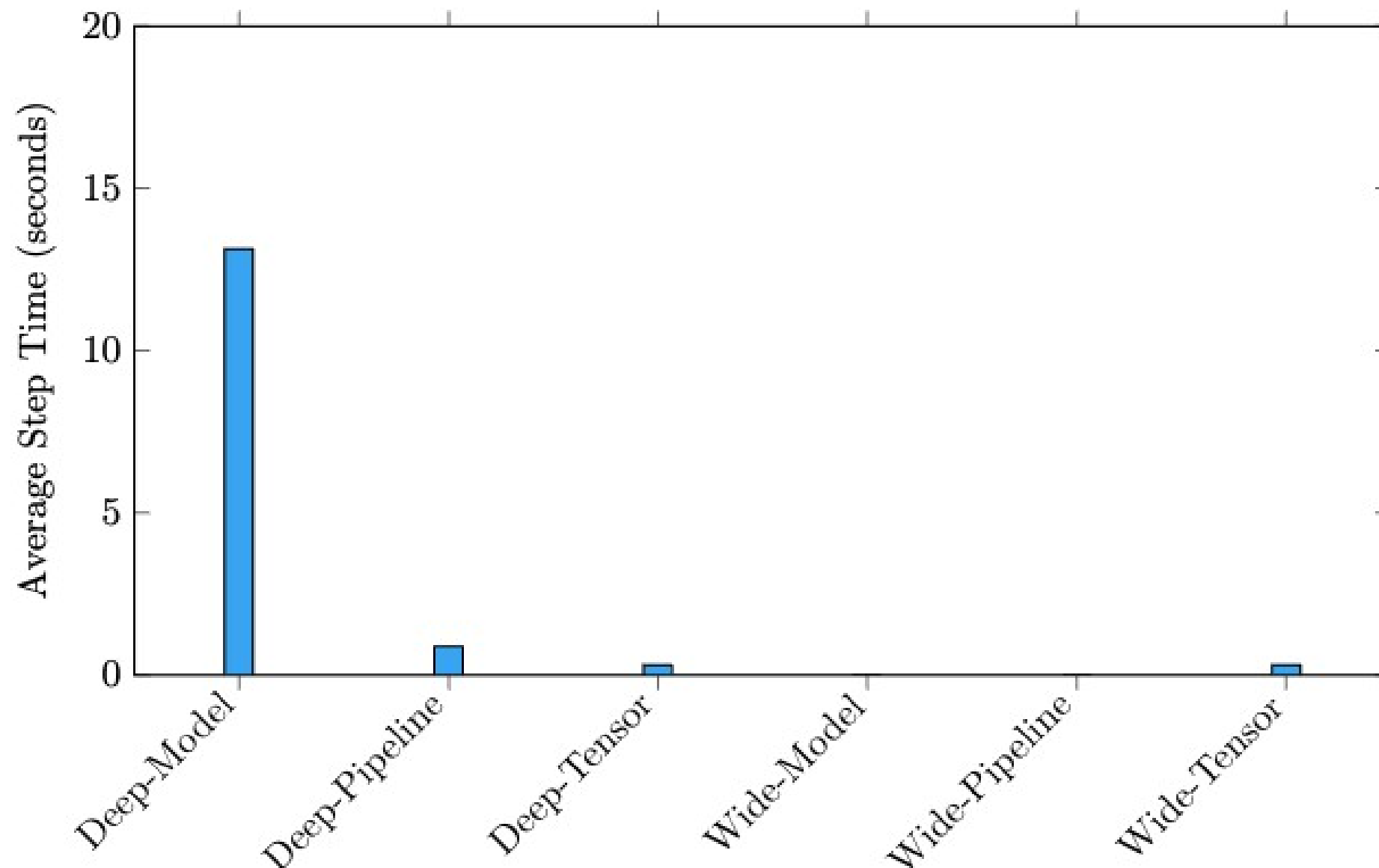
Tensor Parallelism



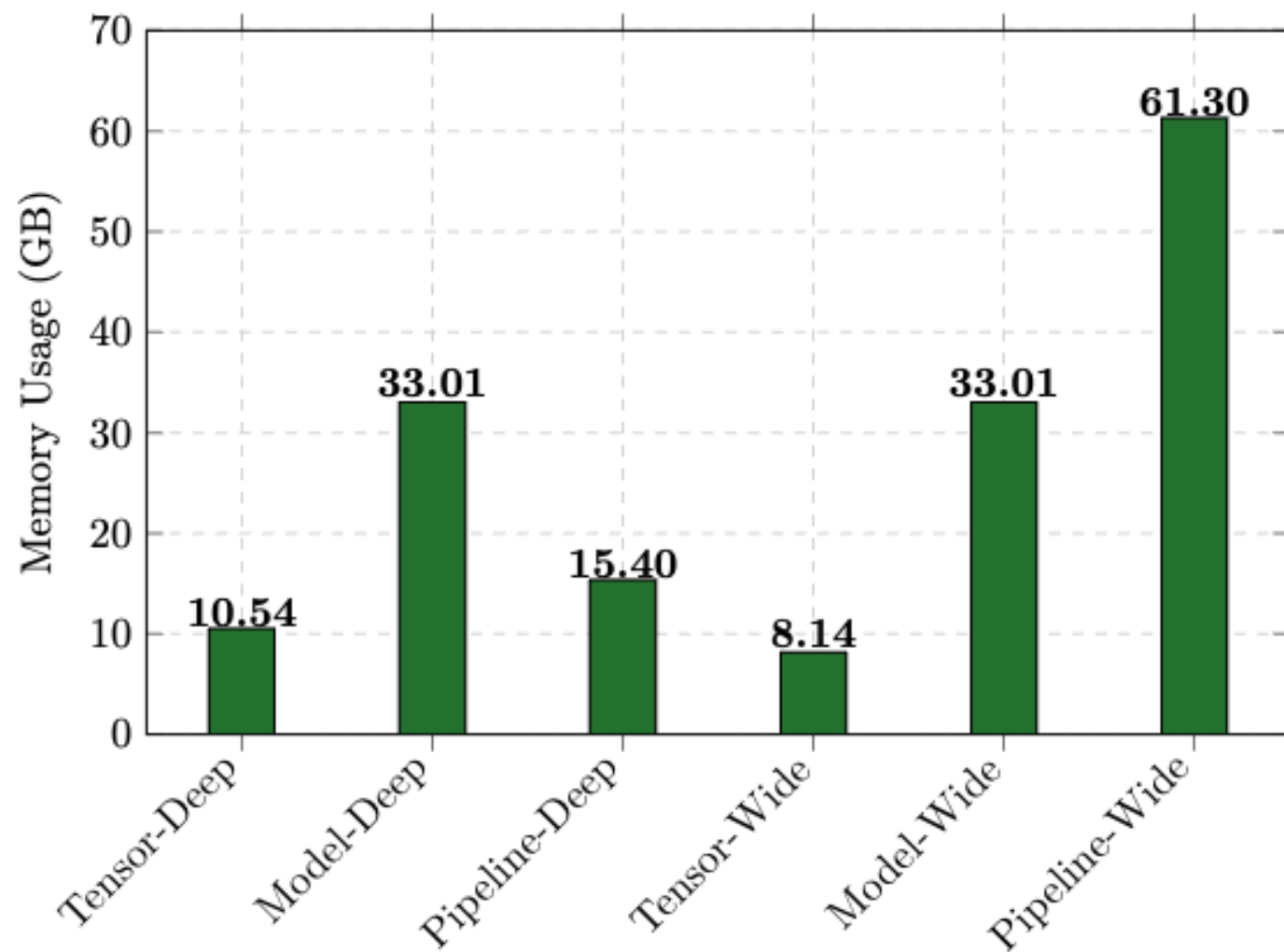
Models tested

- Tested on either 2 or 4 GPUs at a time
 - 1024 embedding dimension x 80 transformer layers; 8 attn head
 - 4096 embedding dimension x 10 transformer layers; 32 attn head
- All transformer-based, since my use is in the same area.
- Next-token prediction on dummy text.
 - **I need to figure out how transformers work – I am confused.*

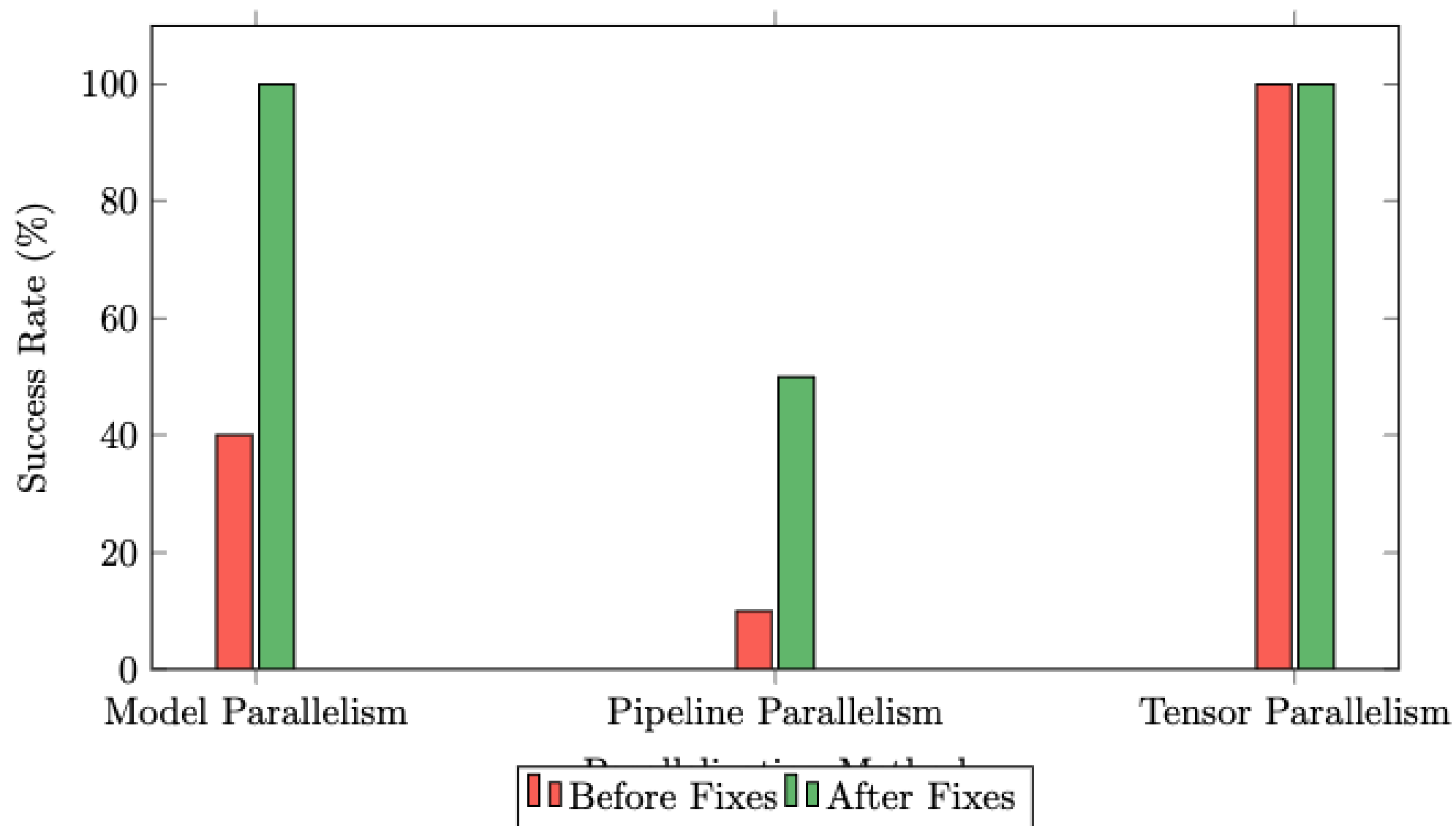
Performance



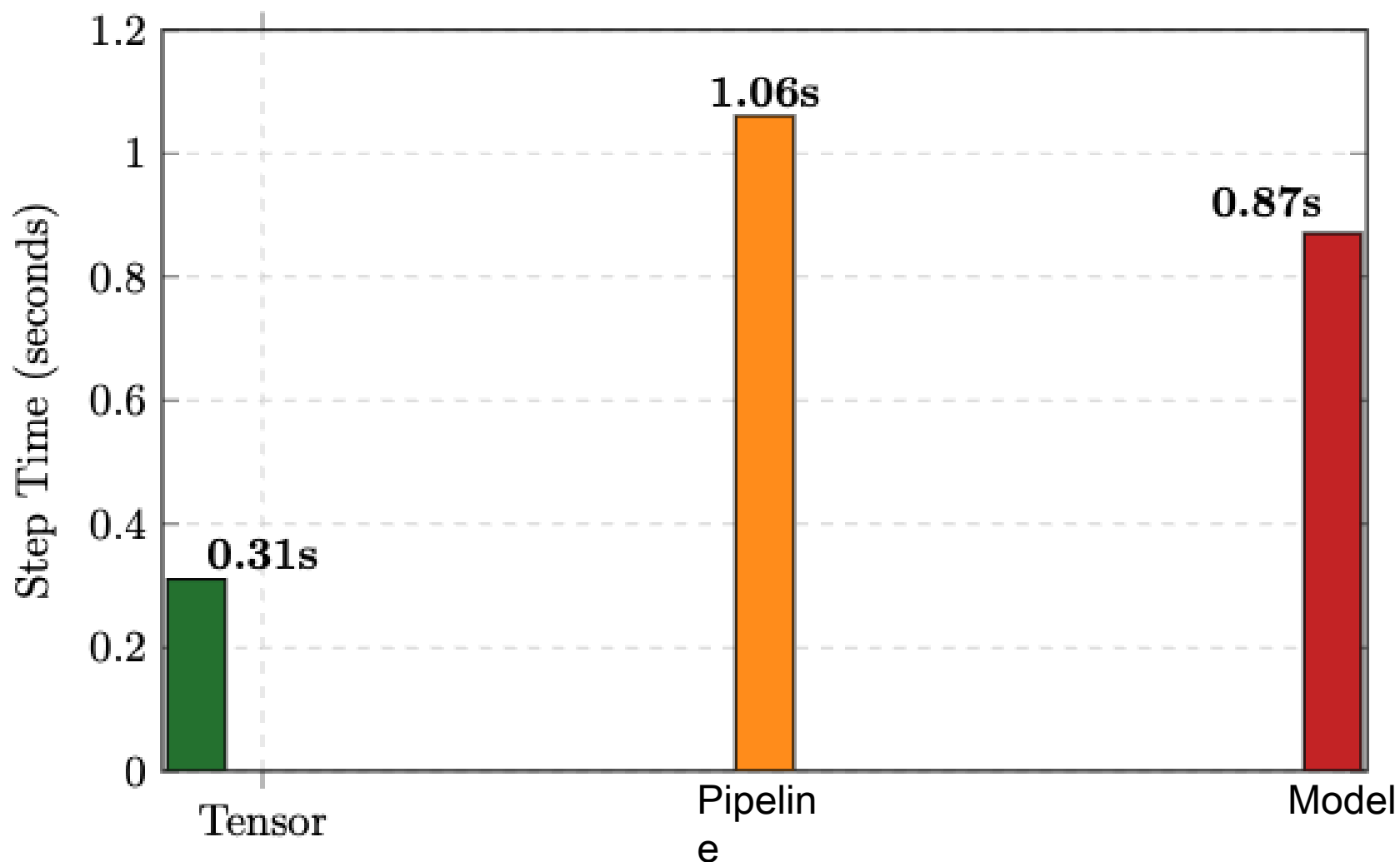
Memory Usage



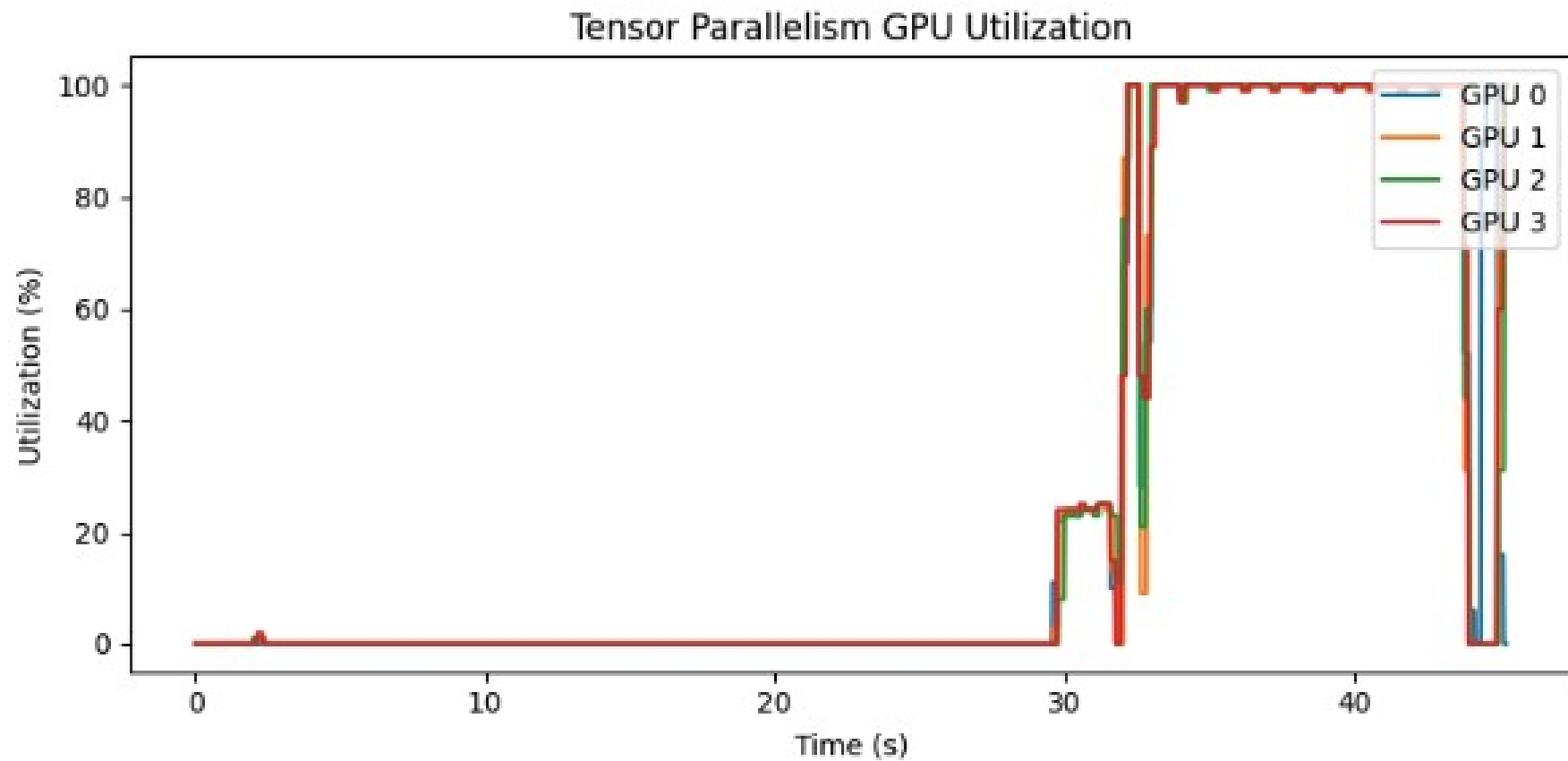
Fixing Errors



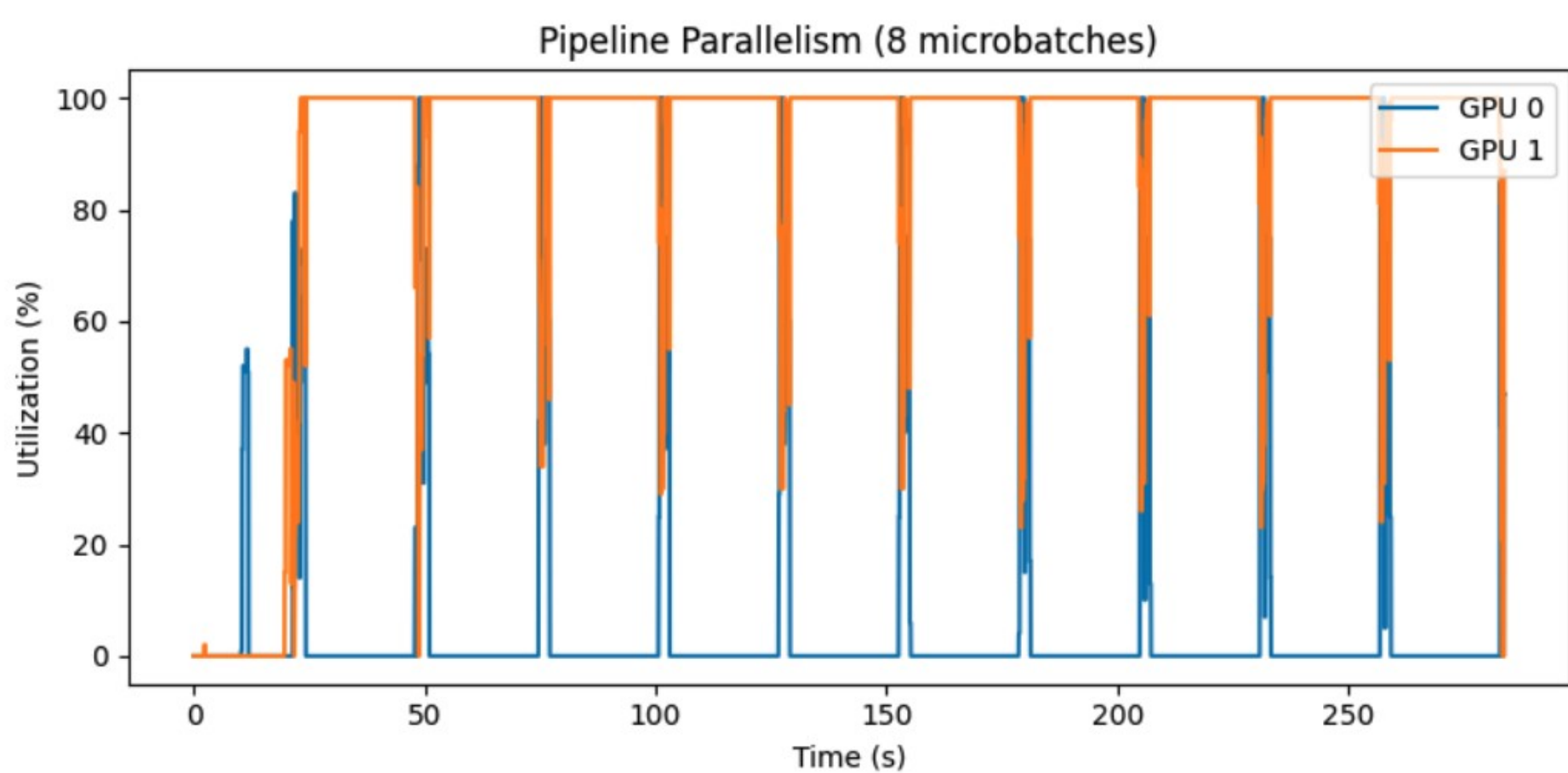
Time per method



GPU Utilization



GPU Utilization



Questions